

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Macroeconomics, Data Structures

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Field interviews with construction workers surfaced two constraints — LIDAR was cost-prohibitive under thin project margins, and workers frequently crossed the camera's line of sight — **chose CV over LIDAR for cost-effectiveness**, helping file a provisional patent for a tripod-mounted 1000fps camera and Jetson Orin GPU capture rig.
- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.

Software Engineering Intern, CACI International Inc. – Virginia

May 2025 – Aug 2025

- Built a full-stack logistics web app for **U.S. Customs field officers** using Angular, Spring Boot, and PostgreSQL.
- Expanded database test coverage **by 20%**, enhancing system stability and enabling more efficient bug detection.
- Ramped onto a large, unfamiliar Spring Boot codebase fast enough to **pick up frontend bug tickets and a backend bug-fix story**, contributing through sprint planning, standups, and code reviews in an Agile team.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.
- Fine-tuned a BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using a custom edit-distance metric.
- Owned the data pipeline for a ~1M-word, ~10K-text raw corpus — preprocessing and alignment work that **had to be right before any model result was credible**.

Projects

PotBot — Civic Reporting Agent, CalHacks Prize Winner

<https://devpost.com/software/potbot>

- Led a team of 3 under a 36-hour hackathon deadline, **prioritizing a working demo over an unfinished feature set** — shipped an agentic pipeline (Llama vision, Playwright) that auto-submits real infrastructure repair requests to SF city government, winning CalHacks' sponsor prize.

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3)

github.com/FavouritePlayer/ant_sim

- Validated a leg-damage-specialist policy across multiple seeds and found it didn't transfer to other legs (100% fall rate) — **killed that direction rather than ship an unsupported claim** — then built a compound policy router that beat any single policy (836 vs. 497 mean reward) across a 4-leg $\times$ 10-seed evaluation sweep.
- Terrain-adapted policy, validated across 10 matched seeds, **achieved 2.1x the flat-baseline reward** (893 vs. 424) and 40% lower fall rate on unseen terrain — the approach that shipped.

DealScout — Resale Arbitrage Agent (LangGraph, Playwright)

github.com/FavouritePlayer/dealscout-cloud

- Redesigned the demo's cloud infrastructure around a hard \$10 budget — **chose a lightweight self-hosted Kubernetes distribution over managed EKS to avoid ~\$73/month in control-plane cost** — while still deploying a real 4-service system (agent, scraper, vector DB, frontend) on AWS via Terraform.

Context Custodian — Workspace Hygiene Agent (FastAPI)

github.com/FavouritePlayer/ContextCustodian

- Built a FastAPI agent that compacts and cleans an agent-first workspace for downstream agents to ingest — a three-stage prompt-injection detection pipeline (regex, vector similarity, LLM adjudication) with a fail-open fallback, where **every fix is a reversible move, never a delete**.

Skills

Languages: Python, Java, JavaScript, SQL

Technologies: React.js, Angular, Spring Boot, Flutter, AWS, Kubernetes, PyTorch, LangGraph, FastAPI, Playwright, Git